Calculus I Assignment 6

1. Find dy/dx by implicit differentiation.

(a) $6x^4 + y^5 = 2x$

(b) $x^2 - 4xy + y^2 = 4$

(c) $x^4 + x^2y^2 + y^3 = 5$

(d) $\dfrac{x^2}{x+y} = y^2 + 1$

(e) $xe^y = x - y$

(f) $\sin x + \cos y = 2x - 3y$

(g) $\sin(xy) = \cos(x + y)$

(h) $xy = \sqrt{x^2 + y^2}$

2. Find an equation of the tangent line to the curve at the given point.

(a) $x^{2/3} + y^{2/3} = 4, \quad (-3\sqrt{3}, 1)$

(b) $x^2 - xy - y^2 = 1, \quad (2, 1)$

(c) $x^2 + y^2 = (2x^2 + 2y^2 - x)^2, \quad (0, \dfrac{1}{2})$

3. Find the points on the lemniscate $(2(x^2 + y^2)^2 = 25(x^2 - y^2))$ where the tangent is horizontal.

4. Show that the sum of the x- and y-intercepts of any tangent line to the curve $\sqrt{x} + \sqrt{y} = \sqrt{c}$ is equal to c.

5. Find all local and absolute extrema for the function $f(x) = \dfrac{x-2}{x^2+1}$ on the interval [-1,2].

6. Show that $f(x) = x^3 + bx^2 + cx + d$ has both a local maximum and a local minimum if $c < 0$.

7. For the family of functions $f(x) = x^4 + cx^2 + 1$, find all local extrema (your answer will depend on the value of the constant c)

8. Find all critical numbers and use the First Derivative Test to classify each as the location of a local maximum, local minimum or neither.

$$y = \arcsin(1 - \frac{1}{x^2})$$

Bonus Questions

1. (a) Show that for $x > 0$,

$$e^x > 1 + x$$

(b) Show that for $x > 0$,

$$e^x > 1 + x + \frac{x^2}{2}$$

(c) Show that for $x \in \mathbb{R}$,

$$|\sin x| \leq |x|$$

(d) Show that for any real numbers u and v,

$$|\cos u - \cos v| \leq |u - v|$$

END